

CP020003

ARTIFICIAL INTELLIGENCE



Khon Kaen University · 2026

Week 08

Modern AI Foundations

Plus: a full recap of Weeks 1–7 and guidance for the Midterm Exam



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https://github.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1

Two Parts, One Goal: Ready for What's Next

1 Part 1 — The Modern AI Landscape

- Multimodal AI — what it is
- The LLM concept, in plain terms
- Foundation models — an up-to-date map
- Agentic AI & AI agents
- Tools of the trade: Hugging Face & PyTorch
- Fine-tuning, LoRA, and RAG

2 Part 2 — Midterm Recap & Exam Guide

- Weeks 1–2: Intro to AI & Feature Engineering
- Weeks 3–4: Supervised Learning & DNA Classification
- Weeks 5–6: Unsupervised Learning & Recommenders
- Week 7: Association Rule Mining
- How the midterm is structured
- How to study & what you may bring

From Traditional ML to Modern AI

Weeks 1–7 focused on classical machine learning: engineered features and structured, tabular data. Modern AI shifts to large models trained directly on massive, unstructured data — text, images, and audio.



This week previews Stage 3 and 4 — you will return to full depth after the midterm.

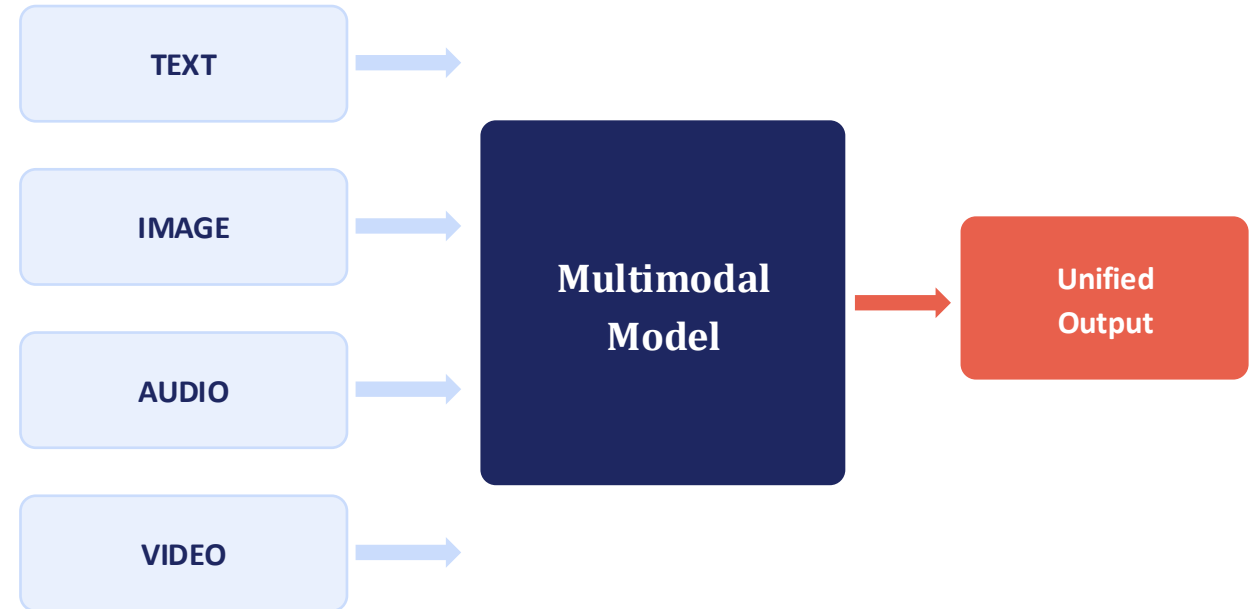
Multimodal AI

Definition

An AI system that can understand and/or generate more than one type of data — text, images, audio, video — and reason across them together, not as separate models bolted together.

Example

Show a model a photo of your fridge and ask, “What can I cook tonight?” — it reads the image and replies in text with a recipe. Or: describe a scene in words and the model generates an image of it.



The LLM Concept

Definition

A Large Language Model (LLM) is a neural network — almost always a Transformer — trained on massive amounts of text to predict the next token (word piece) in a sequence. From this simple task, broad language understanding and generation ability emerges.

Example: “The cat sat on the ____” → the model predicts *“mat”* because that continuation is most probable given everything it learned.

- **Tokens** Chunks of text the model reads/predicts one at a time
- **Parameters** The model's learned weights — billions, for modern LLMs
- **Self-Attention** Lets every token weigh every other token when reasoning

Right: how a base LLM becomes a helpful assistant

Pre-training

Next-token prediction
on huge, general text

Fine-tuning

Trained on curated
instruction examples

Alignment (RLHF)

Tuned to be helpful,
harmless & honest

Deployed Assistant

The chat model
you interact with

Foundation Models — A Snapshot

A foundation model is a single large model pretrained on broad data, then adapted (via prompting, fine-tuning, or RAG) to many downstream tasks. New versions ship constantly — treat this as a snapshot, not a fixed list.

Family	Organization	Known for
GPT (GPT-4 / GPT-4o and later)	OpenAI	General reasoning, multimodal input
Claude (Sonnet / Opus / Haiku)	Anthropic	Long-context reasoning, coding, safety focus
Gemini	Google DeepMind	Native multimodality, very long context
Llama	Meta	Open-weight, widely fine-tuned by others
Mistral	Mistral AI	Efficient, open-weight models

Tip: for the exam, know what a foundation model is and why one model can serve many tasks — you will not be tested on brand names.

Agentic AI: What Is an AI Agent?

Definition: An AI agent is a system built around a foundation model that can **perceive** context, **plan/reason** about what to do, **act** (often by using tools), and **observe** the result — repeating this loop with some autonomy until a goal is reached.

A chatbot answers once. An agent keeps working — in a loop — until the goal is done.

Goal

The instruction or objective it's working toward

Reasoning

The LLM core: plans steps, decides what to do next

Tools

APIs, code execution, web search, files it can call

Memory

Context / history kept across steps of the task

The Agent Loop, in Action



Example: Coding Agent

- Task: “Fix the failing test.”
- Observe: reads the code and the error message
- Think: forms a hypothesis for the fix
- Act: edits the file, re-runs the tests, repeats

Example: Travel-Booking Agent

- Task: “Book me a flight to Chiang Mai on Friday.”
- Observe: checks calendar & budget context
- Act: searches flights via a tool/API, compares options
- Asks for confirmation, then completes the booking

Hugging Face: The Model Hub & Toolkit

Definition

Hugging Face is an open platform and Python ecosystem for sharing and running machine learning models, datasets, and demos. It has become the default place to find and use pretrained models — especially for NLP and, increasingly, all modalities.

- **Hub** Repository of 100,000s of models & datasets
- **transformers** Python library with one common API to load & run models
- **datasets** Library for loading & processing training/eval data
- **Spaces** Hosted demo apps built with a model, shareable via a link

Sample: sentiment analysis in 3 lines

```
from transformers import pipeline

clf = pipeline("sentiment-analysis")
clf("This lecture is great!")

# -> [{'label': 'POSITIVE',
#       'score': 0.999}]
```

PyTorch: The Framework Underneath

Definition

PyTorch is an open-source deep learning framework (originally from Meta AI) built around tensors and automatic differentiation. It is the dominant framework used to build and train most modern foundation models, including the ones behind Hugging Face.

- **Tensor** A multi-dimensional array (like NumPy) that can run on a GPU
- **nn.Module** The building block used to define layers & models
- **Autograd** Automatically computes gradients needed for training

Sample: a tensor and a tiny linear layer

```
import torch
import torch.nn as nn

x = torch.tensor([1.0, 2.0, 3.0])
layer = nn.Linear(3, 1)
y = layer(x)           # forward pass
y.backward()           # autograd!
```

Fine-Tuning a Foundation Model

Definition: Taking a pretrained foundation model and training it further on a smaller, task- or domain-specific dataset, so it specializes for that use case — instead of training a new model from scratch.

Full Fine-Tuning

- Updates all of the model's weights
- Needs the most compute, memory & data
- Produces a full copy of the model per task

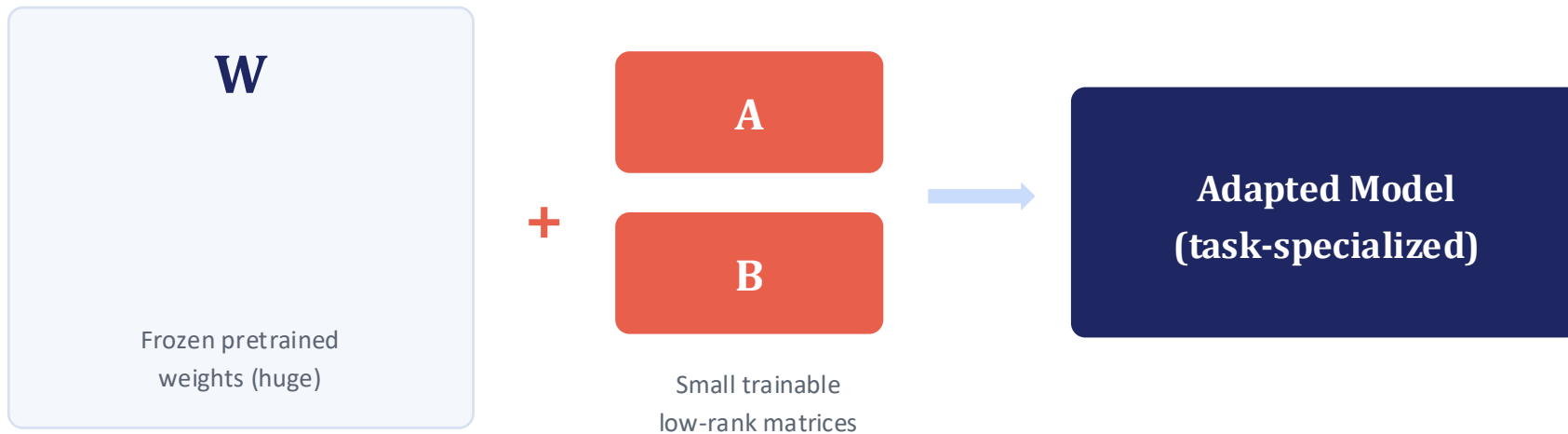
Parameter-Efficient Fine-Tuning (PEFT)

- Freezes most weights, trains only a small added part
- Far cheaper & faster — e.g. LoRA (next slide)
- Small, swappable adapters instead of a full new model

Example: *fine-tuning a general LLM on a Thai bank's customer-service transcripts so it answers in the company's tone and follows its actual policies.*

LoRA — Low-Rank Adaptation

Definition: A PEFT technique that freezes the original weight matrix and injects a pair of small, trainable “low-rank” matrices next to it. Only those small matrices are trained — cutting the compute and memory needed to fine-tune by orders of magnitude.



- Much cheaper & faster than full fine-tuning
- Different LoRA adapters can be swapped on top of the same frozen base model for different tasks
- Often matches full fine-tuning performance for many tasks

RAG — Retrieval-Augmented Generation

Definition: A technique that pairs an LLM with a retrieval step: at query time, the system searches a knowledge base for relevant text, then feeds those passages into the prompt before the model answers — grounding the response in real, current facts instead of only the model's trained memory.



Example: an internal HR chatbot that retrieves the exact policy PDF section before answering an employee's question — instead of the LLM guessing from general training data.

Nine New Terms in One Table

Term	In one line
Multimodal AI	One model that understands/generates several data types together
LLM	Transformer trained to predict the next token; broad language ability emerges
Foundation Model	A large pretrained model reused across many downstream tasks
Agentic AI / AI Agent	A system that observes, plans, acts via tools, and repeats toward a goal
Hugging Face	Hub + libraries for finding and running pretrained models & datasets
PyTorch	The deep learning framework (tensors + autograd) most models are built on
Fine-tuning	Further training a pretrained model on a smaller, specific dataset
LoRA	A cheap fine-tuning method: train small added matrices, freeze the rest
RAG	Retrieve relevant text first, then let the LLM answer using that context

PART 2

Midterm Recap: Weeks 1–7

A fast pass through everything you've learned so far, plus how to approach next week's exam.

Weeks 1–2: Foundations & Feature Engineering

Week 1 — Intro to AI

- AI vs. traditional programming: learning from data vs. hand-written rules
- Types of ML: supervised, unsupervised, reinforcement learning
- Six classical models: Decision Tree, Random Forest, XGBoost, Logistic Regression, SVM, Neural Network / MLP

Week 2 — Feature Engineering

- Turning raw data (Spotify audio attributes) into model-ready features
- One-Hot Encoding: unordered categories → separate binary columns
- Label / Ordinal Encoding: categories with a natural order → one numeric column
- Good features often matter more than the choice of model

Weeks 3–4: Supervised Learning & DNA Classification

Week 3 — Supervised Pipeline

- Train/test split; Stratified split preserves class ratios when classes are imbalanced
- Overfitting: model memorizes training data, fails to generalize
- Precision vs. Recall, plus Accuracy, F1 and ROC-AUC
- Curse of dimensionality: too many features can hurt distance-based models

Week 4 — DNA Classification

- Applying supervised classification to biological sequence data
- Representing sequence data as usable features for a model
- Convolution (CNN): a filter slides across input data to detect local patterns
- Choosing a model based on the structure of the data, not habit

Weeks 5–6: Unsupervised Learning & Recommenders

Week 5 — Unsupervised Learning

- K-Means & the Centroid: the (moving) center point of each cluster
- Evaluating without labels: Silhouette, Davies-Bouldin, Calinski-Harabasz
- When metrics disagree on the best K, choose using business context
- DBSCAN as an alternative for non-spherical, density-based clusters
- RFM features (Recency, Frequency, Monetary) for customer segmentation

Week 6 — Recommender Systems

- Content-based vs. Collaborative Filtering vs. Hybrid approaches
- Cold-Start Problem: new users and new items have no history yet
- Offline metrics (e.g. Precision@K) to check Top-N quality before launch
- Risks of relying on collaborative filtering alone: diversity & fairness

Week 7: Association Rule Mining

- Transactions & itemsets: what items were purchased together
- Support, Confidence, and Lift measure how strong a rule $\{A\} \Rightarrow \{B\}$ is
- Leverage & Conviction add further nuance to rule strength
- Confidence is not symmetric: $\text{Confidence}(A \Rightarrow B)$ usually $\neq \text{Confidence}(B \Rightarrow A)$
- Business use: combo pricing, product placement, promotions

Formulas to know cold

$$\text{Support}(A) = \text{count}(A) / N$$

$$\text{Support}(A, B) = \text{count}(A \text{ and } B) / N$$

$$\text{Confidence}(A \Rightarrow B) = \text{Support}(A, B) / \text{Support}(A)$$

$$\text{Lift}(A \Rightarrow B) = \text{Confidence}(A \Rightarrow B) / \text{Support}(B)$$

$\text{Lift} > 1 \rightarrow A \text{ and } B \text{ occur together}$
more than by chance

How the Midterm Is Structured

What to expect

- Term & concept questions: define a term concisely and give one example
- Case-study questions: apply a full pipeline to a realistic business scenario
- Topics span all 7 weeks — supervised design, clustering & segmentation, recommenders, and hand-calculated association rules
- You will be asked to justify decisions, not just name them

What you may bring

- Your own notes, printed slides, or the course textbook — allowed
- No internet access during the exam
- No computer or laptop during the exam
- No mobile phone during the exam

How to Study This Week

1

Rebuild the glossary

For every bolded term across Weeks 1–7, write a one-line definition plus one concrete example — from memory, then check it.

2

Redo the hand math

Practice Support, Confidence, Lift, Leverage and Conviction on a small transaction table until the arithmetic is automatic.

3

Practice the “why”

For each case-study topic, rehearse: Decision → Reasoning → Trade-off. Graders reward justification, not just the right label.

4

Pack only what's allowed

Printed notes or the textbook — yes. Phone, laptop, and internet — no. Prepare your paper materials in advance.

You've built the full picture — classical ML through modern AI.

Good luck on the midterm —
review the recap, redo the math by hand, and bring only what's allowed.